Uncertainty Communication in LLMs: Can Fine-tuning Improve Calibration in Question Answering?

Bachelor's Thesis

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Abstract

Large Language Models (LLMs) are known to produce text that is not factually correct. Errors, known as hallucinations, often arise from internal model uncertainty during generation. In the context of question-answering, different methods of measuring uncertainty have been developed. However, researchers are still exploring techniques for communicating it to the user. In this study, we demonstrate the potential of a fine-tuned solution to produce human-like expression of confidence during answer generation. We achieve this by producing datasets that represent different levels of confidence for the generated answers of the baseline Gemma 3 models of different sizes and using them for additional training. We demonstrate that fine-tuning for uncertainty expression does not reliably improve calibration; however, fine-tuning shows more consistent improvement in discriminability. We also attempt to evaluate the effects of domain similarity on results of fine-tuning by isolating question topics in training and testing; however, we were unable to observe any meaningful relationship between domain proximity and changes in calibration and discriminability after fine-tuning. Our work demonstrates that the development of such models is worth investigating further, especially given the recent growth of interest in implementing these systems in professional environments and personal use. Human-understandable expression of uncertainty can be especially beneficial for highly sensitive fields, where misinformed decisions can have a negative impact on others.

The project repository can be found here.

1 Introduction

Over the last couple of years, LLMs have outgrown their novelty status and have become part of the daily lives of millions of users (Malik, 2026). They have demonstrated strong capabilities across a variety of text-based tasks, including question answering, summarization, and programming. However, the adoption of Large Language Models is held back due to the well-known challenge of detecting hallucinations: there are limited reliable mechanisms for end users to distinguish between valid and fabricated claims (Farquhar et al., 2024).

In addition, the general notion of chatbot agents is associated with anthropomorphism and imitating human interaction patterns (Alabed et al., 2022). However, it also increases user trust in the factual accuracy of model outputs, particularly when responses are perceived as confident.

Hallucinations are often linked to predictive uncertainty of the output in Language Models (Xiao and Wang, 2021). Prior work suggests that, in a free-form simple question-answering context, uncertainty can be estimated (Kadavath et al., 2022; Chen and Mueller, 2023; Farquhar et al., 2024). Existing approaches, as described in a survey by Huang et al. (2024), span probabilistic, ensemble-based, and information-theoretic methods [1].

Surveyed methods are mostly focused on estimating uncertainty of LLMs in a controlled experiment setting with multiple generations, access to weights, or using an output format that detracts from the anthropomorphism that makes LLMs accessible (Xiong et al., 2024).

To address the issue of unwarranted confidence in LLMs, Ulmer et al. (2026) proposes the term *anthropomimetic uncertainty*, which refers to the idea that language models should adhere to human uncertainty communication. The study explores different methods of eliciting uncertainty from language models and formats (registers) of uncertainty communication in LLMs. The authors point out the domination of numerical expressions as the method of expressing uncertainty used in the field.

In addition, according to Ulmer et al. (2026), prompting remains the main method of eliciting expression of uncertainty from the model. While suitable for black-box models, this method might not be appropriate for the calibration of uncertainty expression in language models as suggested by Yona et al. (2024).

Supervised fine-tuning (SFT) is used in the field as one of the methods for eliciting uncertainty communication. But, more often than not, the chosen method of uncertainty expression is numeric or verbalized confidence in the decoded output of the language model. (Kadavath et al., 2022; Li et al., 2025; Wang et al.,

[1] See Section 2.1 for definition of uncertainty and confidence

2024).

Numeric methods are readable and explicit. However, they do not fully replicate the natural expression of uncertainty (Tao et al., 2025b). Another research approach for uncertainty communication for LLMs is *Linguistic Confidence* (LC). Similar to human speech, hedged language can signal uncertainty in LLM output via the use of factive verbs, modal verbs, adverbs, and complex expressions (e.g., "I am not an expert, but you might want to try restarting your computer").

Tao et al. (2025b) proposes a fine-tuning pipeline for uncertainty communication using Linguistic Uncertainty rather than numerical expressions. In their work, the answer accuracy of the baseline model, measured as Semantic Uncertainty (see Related Work), is used to construct a fine-tuning dataset in the question answering setting.

In this work, we evaluate the feasibility of fine-tuning for enabling human-like uncertainty expression in question-answering settings for LLMs at inference time. The goal is to enable better uncertainty expression during answer generation and to produce more calibrated outputs. In particular, we reconstruct the pipeline developed in Tao et al. (2025b) and evaluate its robustness for alternative model sizes using Gemma 3. We further evaluate performance for the out-of-distribution (OOD) setting by isolating several domains in training and testing.

We focus on the effect of fine-tuning on calibration of uncertainty communication and answer accuracy, as well as the utility of communicated uncertainty in differentiating correctly and incorrectly answered (i.e, discriminability).

Our research questions are as follows:

1. Does the size of the model affect the results of fine-tuning for uncertainty expression?

2. If the model is fine-tuned on answers from one domain, can we see an improvement in expressing uncertainty for out-of-distribution questions?

3. Does the improvement of uncertainty calibration correlate with the semantic similarity between the fine-tuning and evaluation domains?

2 Related Work

In this section, we review prior work related to uncertainty estimation, uncertainty communication, and calibration in LLMs, with a focus on calibration methods and linguistic uncertainty expression.

The idea of calibration is discussed in Guo et al. (2017), where it is defined as the alignment between predicted probability estimates and the true likelihood of correctness. The original work focuses on calibration in models that output explicit label probabilities, which makes evaluation of calibration relatively straightforward.

In Large Language Models, model output can be considered in two ways: the token probability distribution of the output sequence and the meaning of the decoded text. As reflected upon in the field, sequence likelihoods do not directly correspond with the correctness likelihoods or perceived answer confidence in the question answering context. Thus, in order to achieve alignment between answer correctness likelihood and uncertainty expression, both model uncertainty and output text uncertainty need to be defined in a way that considers the semantic meaning of the output. (Kadavath et al., 2022)

As mentioned in Farquhar et al. (2024), token probabilities express the uncertainty of the model over phrasings that do not change the meaning of an output. Consequently, the token likelihood of an output sequence is affected by both the semantic meaning of the answer and the form of the answer. This prevents token-level probabilities from directly corresponding to the probability of factual correctness or to linguistically expressed uncertainty. As a result, specialized methods for uncertainty estimation in LLMs are needed.

2.1 Definition of Uncertainty and Confidence

We define uncertainty and confidence in relation to large language models in the following way:

Confidence is used to communicate the likelihood of a specific answer, and it is used to represent how strongly the model supports the final selection.

Example: If the same model answers, "The capital of Australia is definitely Sydney," it is expressing high confidence in a specific response, even though that answer is false. This shows how confidence and correctness are not the same thing.

On the other hand, **uncertainty** refers to the properties of the distribution of potential sequences that the model could generate.

Example: A model is asked to complete the sentence, "The capital of Australia is …" and it considers both "Canberra" and "Sydney" as plausible continuations, even though only one is correct.

Although related, the two concepts are not equivalent (Huang et al., 2024; Tao et al., 2025b).

2.2 Uncertainty Estimation in Language models

Uncertainty estimation in large language models can be categorized by access requirements into black-box methods and white-box methods. White-box methods require access to model output distributions, whereas black-box methods allow estimation of uncertainty only with the use of model outputs.

From the perspective of signal type, the methods can use token probability, sampling variability, semantic grouping, and natural language as input.

2.3 Token probability methods

The following methods use the token-level probability distributions of the model output as the main input for estimating uncertainty. They are grouped together because they rely on surface-level probability distributions rather than the semantic meaning of the model output.

Predictive Entropy

Kadavath et al. (2022) utilizes predictive entropy over the answer distribution as a baseline method of estimating the accuracy of the model in QA. In their work, they demonstrate that the average entropy of correctly answered questions is lower than that of those answered incorrectly.

Despite that, the authors suggest that questions with a diverse set of correct answers do not allow predictive entropy to be used reliably as an estimation method for model uncertainty. Researchers also note the negative trend of the usefulness of predictive entropy for discriminability[2] with increasing model size for some tasks. Kadavath et al. (2022) hypothesizes that this is due to larger models being capable of giving more diverse *correct* answers.

This finding further emphasizes the distinction between token-level uncertainty and semantic uncertainty.

P(True)

Kadavath et al. (2022) also explores the token probability of answer sequences as a confidence predictor in multiple-choice and true-or-false question answering. In practice, this means that questions are accompanied by a prompt with instructions for the model to either choose from (A, B, C, D) or to return True or False.

The assigned probability of a specific answer option in the output layer is then treated as a proxy for correctness likelihood. This value is referred to as P(True) and can be used to estimate the uncertainty of the model in question-answering, albeit in limited settings.

The study highlights the discrepancy between uncertainty estimation as evaluated by external correctness labels or sequence token distributions and the model's internal representation of its knowledge.

2.4 Semantic Grouping Methods

This family of methods is based on the consistency of the meaning of the output sequences rather than token-level probabilities. Semantic Equivalence can be evaluated manually, using a Large Language model, or with the help of a natural language inference (NLI) classification system.

Semantic Entropy

Kuhn et al. (2023) proposes semantic entropy as a method of uncertainty estimation that addresses the gap between model and output uncertainty. In order to distinguish between the "meaning space" and the "token space", the model response samples are clustered into meaning classes using NLI classification. Entropy is calculated across the distribution of semantic clusters (i.e., answers equivalent in meaning) rather than individual sequences.

Researchers report improved discriminability compared to predictive entropy-based uncertainty measures, as well as a positive trend of AUROC with a larger model size.

Semantic Uncertainty (SU)

In Kuhn et al. (2023), it was shown that for Semantic Uncertainty, correctly answered questions have a lower spread of different response sequences on average. The authors state that using distinct answer counts can yield reasonable results as a measure of uncertainty even in the absence of entropy computations.

Tao et al. (2025b) utilizes this method to evaluate uncertainty in the base model prior to supervised fine-tuning. We use it since it takes into account semantic meaning in estimating model uncertainty rather than surface-level token distributions. In addition, it allows uncertainty estimation in black-box models, enabling comparisons across both open- and closed-weight models.

2.5 Self-consistency Methods

We refer to the methods that estimate uncertainty from variability across sampled model outputs as self-consistency methods. In these methods, uncertainty is derived from the disagreement between individual samples.

We distinguish self-consistency methods from semantic grouping methods based on grouping criteria. Self-consistency aggregates uncertainty over output sequences without considering the semantic meaning of the answers. In contrast, semantic grouping methods cluster outputs by meaning, thereby estimating uncertainty in a semantic-aware way. Both approaches rely on sampling model outputs but differ in the representations of the output that are used for estimation.

Variability in generated samples can be introduced by adjusting the decoding strategy (top_k, top_p, temperature, seed; for our selected approach, see Sampling). Samples can also be generated using query perturbations such as paraphrasing or prompt variations (Chen and Mueller, 2023) or methods that augment the model during inference, such as Monte-Carlo Dropout (Gal and Ghahramani, 2016).

Wang et al. (2022) proposes the idea of self-consistency in the Chain-of-Thought prompting setting for the tasks where an exact discrete answer is required, such as arithmetic reasoning, math problem solving, and knowledge reasoning.

2.6 Language-Based Methods

We identify the methods that use model output to express uncertainty as Language-Based, as they are not based on token sequence likelihood estimation, but rather exclusively on the explicit meaning of the decoded output.

[2] As measured by AUROC. See Methods.

Verbalized Confidence

Prior work on verbalized confidence studies whether language models can communicate calibrated uncertainty through natural language expressions rather than token probabilities. Lin et al. (2022) shows that models can be prompted to output confidence in categorical labels and numerical form.

Tian et al. (2023) tests different methods for eliciting verbalized confidence. Authors show that LLMs are highly sensitive to prompt formulation. Their results imply that changes in prompting may significantly affect the calibration of uncertainty expression, suggesting that verbalized confidence as a measurement of model uncertainty is unstable.

We differentiate between Verbalized Confidence estimation methods and Linguistic Confidence (see Linguistic Confidence in LLMs) as the former relies on explicit approximation of the confidence level in the output text, whereas the latter aims to convey certainty in a human-like way, with the use of hedged language and specific linguistic constructions to communicate model uncertainty in the decoded output.

Learned Confidence

Evaluation of the confidence level based on the use of hedged language requires mapping verbal expression into numerical confidence scores. LLM prompting or human annotation is often used for the evaluation of the confidence level in the evaluated samples. Researchers in Tao et al. (2025b) propose a learned BERT-based mapper that predicts confidence directly from the inputs.

The choice of the estimation method can have implications for fine-tuning for uncertainty communication. Depending on how uncertainty is defined, the training signals and loss functions may take different shapes. In this work, we adopt SU as proposed by Tao et al. (2025b) to estimate uncertainty and apply it as the supervision signal in our fine-tuning pipeline. The aforementioned BERT-based confidence mapper is used to extract confidence levels of the tested models in order to estimate calibration and discriminability. More details on specific implementations of these methods in our research can be seen in Methods

2.7 Uncertainty Communication/Expression

Linguistic Confidence in LLMs

Some of the methods we discussed, like Verbalized Numerical Uncertainty, are based on the model evaluating its own answer, which brings a separate layer of discussion, whether or not the model has accurately assessed the output. For methods like Semantic Uncertainty, evaluation takes time and a lot of computation. And most importantly, even though most of the approaches show potential for addressing and measuring the problems of hallucination, their interpretation requires additional information, which might ultimately lead to confusion for the user.

As a solution to this issue, some authors propose focusing on evaluating and improving LLMs' abilities to reflect their uncertainty in the answer generation itself (Tao et al., 2025b) (See example in Appendix E). The authors refer to the method of uncertainty communication that follows this concept as Linguistic Confidence (LC), also referred to as Linguistic Verbal Uncertainty (LVU)(Tao et al., 2025a). We will use LC as a term for this method.

LC assumes that the model should output sequences with Words of Estimative Probability (WEPs)(Tang et al., 2026), such as “maybe”, “probably not”, ”definitely”, etc., suggesting to the user that the answer they see might be partially true or even false.

To assess the potential of this method, many studies have evaluated a variety of SOTA models (Huang et al., 2024; Tao et al., 2025b; Tang et al., 2026). As mentioned above, language models tend to be overconfident (Xiong et al., 2024). Therefore, under the standard conditions in the question answering setting, the user would only ask their question, and the model would proceed to return an answer that is not optimized for accurate use of WEPs. This is replicated in many experiments by using an example prompt like:

Example prompt: *Answer the following question using a succinct (at most one sentence) and full answer*. (see also Appendix Appendix C)

To test the actual potential of LC expression, researchers have tested how the outputs change if the prompt specifically asks for the use of hedging language (Yona et al., 2024; Tao et al., 2025b).

Example prompt: *Answer the following question using a succinct (at most one sentence) and full answer. If you are uncertain about your answer to the question, convey this uncertainty linguistically by precisely hedging this answer.* (see Appendix C)

We will refer to this method as LC+. Early testing has shown that prompting alone was not enough to produce significant improvements in model calibration (Yona et al., 2024), which is likely due to the models failing to capture the use of WEPs accurately or failing to align them with human perception (Tang et al., 2026).

The same researchers state that, for cases where certainty is very high, models perform significantly better in aligning with human interpretation. This further supports the idea that LLMs tend to generate overconfident answers because they can reflect high confidence more accurately. One of their arguments as to why this happens is that the spectrum of estimative words is not equally represented in training data.

Furthermore, words associated with high uncertainty levels, such as “maybe” and ”likely”, are more ambiguous (Tang et al., 2026). Usually, they depend on the context implied by the speaker, and this makes

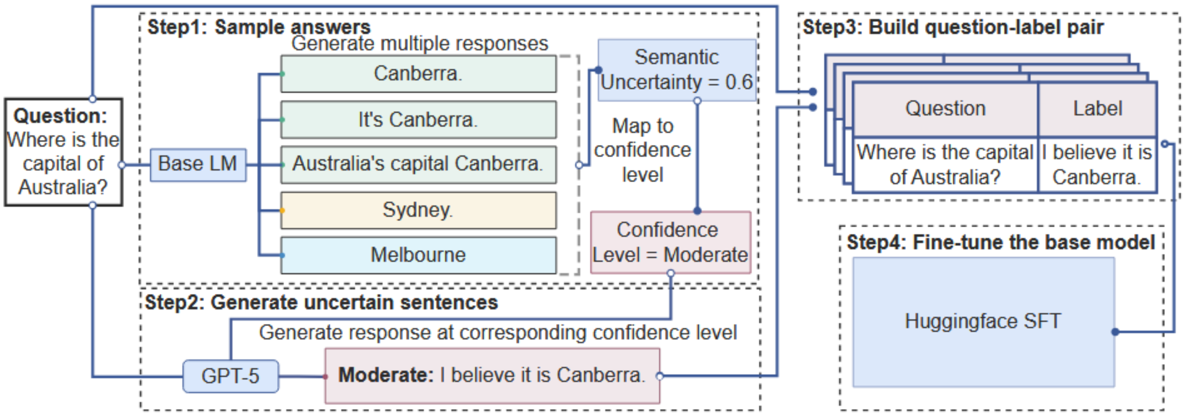


Figure 1: Pipeline visualization (Tao et al., 2025b)

their accurate usage harder to comprehend for LLMs. So what happens is that during training, the ability to distinguish between the lower levels of confidence is traded off for better generalization. This is one reason why we emphasize fine-tuning with equal representation of confidence levels later in this work.

Even though initial testing of LC+ did not show a significant improvement in uncertainty communication, recent tests of the methods present a new trend, where LC+ methods are showing promising performance in communicating confidence, especially when they have access to reasoning. Tao et al. (2025a) presents a broad analysis that compares LC+, VNC, and Token Probability-Based Uncertainty. The main contributions of that work are the comparison of methods and the evaluation of baseline models of different sizes and their respective instruction-tuned versions. Their results show that larger models generally achieve both higher calibration and discriminability, and reasoning may play a major role in reducing overconfident answers (Tao et al., 2025a).

Improving LC capabilities

Tao et al. (2025a) introduces a framework for improving LLMs' ability to communicate confidence in another paper - Tao et al. (2025b). The goal was to develop a model capable of producing human-like verbalized confidence without explicitly requesting the model to use hedged language in the prompt.

The study focuses on fine-tuning a model to express LC and evaluates its performance in relation to the baseline model. They produced a single Qwen3-8B model that showed improvements in LC results. In our study, we utilize a framework proposed in that paper.

We decided to select a different model family in order to evaluate the robustness of the original idea. We selected Gemma 3 models (Gemma, 2025) that are primarily built for local usage, where constant access to online resources and Retrieval-Augmented Generation (RAG) is not required. Our expectation is that offline LLM usage would be the case, where accurate communication of uncertainty would be most valuable.

The pipeline consists of 4 steps displayed in Figure 1. The proposed framework is initialized by querying a set of questions to the targeted model to produce information about the initial uncertainty level.

The authors proposed SU as the uncertainty metric, since it is one of the most reliable black-box methods according to the literature (Farquhar et al., 2024; Tao et al., 2025b). A single question is asked 10 times, and then the 10 answers are grouped. If the answers are semantically equivalent, they are grouped under the same label as shown in the example in Figure 1.

Then the SU score is calculated and mapped to a categorical Confidence Level label using the scale in the Appendix E, provided by the authors. After that, the most common answer is rephrased by an SOTA LLM according to the confidence label and combined with the original question as an example of input and desired output (see Appendix B). The pairs are combined into a fine-tuning set, and the original model undergoes training that shows how to express uncertainty in question answering.

For the 4th step, which is the fine-tuning of the base model, the original framework uses Low Rank Adaptation (LoRA) (Hu et al., 2021) - a broadly used method initially created for language models that allows faster training compared to full fine-tuning methods.

In full finetuning, the production of a new model for a new custom task or domain is done by training the whole structure of parameters, which is inconvenient when experimenting with strong LLMs. However, LoRA takes into account that there is no need to update all parameters in order to influence the behavior of the model or its task.

In fact, it is beneficial to preserve most of them as they are essential for the model's capability to understand language. LoRA finetuning targets the so-called adapter layers and minimizes the computational cost needed for fine-tuning of the models.

One side benefit of LoRA is that it exhibits stronger regularization effects and lower catastrophic forgetting than full fine-tuning, which can reduce overfitting risks (Shuttleworth et al., 2025). This is important since the fine-tuning sets we use for training are relatively small.

3 Dataset

For our research and to compare with the original results of the fine-tuning framework, we decided to use the datasets used for confidence evaluation in Tao et al. (2025b) - SimpleQA (Wei et al., 2024) and NQ-Open (Lee et al., 2019).

Both are collections of factual questions that are designed to test the model's ability to retrieve information that is captured during training. The purpose of the set is to address hallucination specifically, and to enable this, gold answers are precise and make the evaluation with a generated output straightforward.

We wanted to conduct a fair comparison between the different models, so we selected 200 samples from each of the two datasets and computed scores using all three methods mentioned above: LC (asking for a short answer to a question), LC+ (asking for a short answer to a question with a request for hedging), and LC(SFT) (asking the fine-tuned model for a short answer). The intention for this was to produce some observations that show whether we would replicate the results from the said paper with the Gemma models.

In order to expand our comparison and investigate the transferability of fine-tuning, we designed an experiment using out-of-domain data. An additional benefit of the SimpleQA set is that all questions are assigned to a specific category, which creates the possibility of investigating experiment effects in distinct domains. You can see the domain distribution in Table 1. In NQ-Open, every question available was only labeled as "Misconceptions".

Dataset	Category	Count
SimpleQA	Science and technology	858
	Geography	424
	Sports	368
	Art	550
	Politics	709
	Other	475
	TV shows	293
	Music	341
	History	173
	Video games	135
	Total	**4326**
NQ-Open	Misconceptions	3609

Table 1: Distribution of categories across SimpleQA and NQ-Open datasets.

Science and technology was selected as the domain for the fine-tuning set. This category had enough samples to achieve the recommended number of 200 questions (as recommended in Tao et al. (2025b)) and additional samples for testing. Alongside Science and Technology, the models were tested on Art and Geography, as they would all differ semantically from each other. We picked only 100 test questions from each domain, mainly due to restrictions in computational resources.

For Science and Technology, we made sure that the questions used for fine-tuning are not used in the test set. The same applies to the replication tests mentioned above.

4 Methods

In this work, we mainly focus on the replication of Tao et al. (2025b), but we also provide some contributions related to the efficacy of the framework.

The following section is a detailed overview of the framework, complemented by the changes we decided to make along our replication. Some of them were due to an inconsistency with the logic of the pipeline regarding the measurement of the SU score, and others were purely because of resource limitations.

4.1 Evaluation

As we wanted to replicate the framework as suggested in Tao et al. (2025b), we used the same measuring metrics.

Expected Calibration Error

In this work, we evaluate the calibration of LLMs in expressing uncertainty by comparing predicted linguistic confidence with answer correctness. The idea for this is named as Expected Calibration Error (ECE) and its variants (Yona et al., 2024).

ECE is calculated as the mean of the difference between the average confidence and the answer accuracy across m bins, as formally expressed in Eq. 1

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right| \quad (1)$$

where B_m denotes the set of samples in bin m, n is the total number of samples, acc(B_m) is the empirical accuracy within the bin, and conf(B_m) is the average predicted confidence.

Because we take the absolute value of the difference, this allows us to accurately take into account all possible answer classifications: accurate and confident answer, inaccurate but confident answer, accurate and insecure answer, or inaccurate and insecure. Therefore, lower values of ECE show better alignment between the predicted linguistic confidence of the output and the model accuracy. Following the approach of Tao et al. (2025b), we do not include cases where the model has refused to provide an answer to the question.

AUROC

To evaluate the discriminability of uncertainty estimates, we use the Area Under the Receiver Operating Characteristic curve (AUROC). AUROC measures how well the predictions allow distinguishing between correct and incorrect answers.

We assign a linguistic confidence score to each prediction as well as a correctness label. AUROC is calculated by varying the separation threshold set by the confidence (scaling from 0.0 to 1.0), measuring the true positive and false positive rate between accuracy labels, and obtaining the area under the constructed curve. Higher AUROC indicates better separation between answers that belong to the "Correct" or to the "Incorrect" classes. For the calculations here, we include not-attempted questions.

4.2 Uncertainty Estimation

To calculate the confidence and accuracy needed for the formulas above, we first have to perform an analysis of each answer. Several language models were used for this part.

For confidence extraction, we follow the approach from Tao et al. (2025b), so we use a DistilRoBERTa (Sanh et al., 2020) combined with their linear layer regression head trained on human-annotated data as a confidence mapper. The reason for this was that we aimed to keep our work close to the original framework for better comparison.

According to the evaluation, this solution had the smallest mean squared error (MSE), and it is also cost-efficient. This is why they used it for evaluating the LC in their work.

To evaluate answer correctness, we relied on SOTA LLM grading via the instruction prompt from the same paper (you can also find it in Appendix D). Our choice was to employ GPT-OSS 120B (OpenAI, 2025) and Nemotron 3 Super (NVIDIA, 2025) for this task.

4.3 Improving Linguistic Confidence

Here, we present our fine-tuning process. For our experiments, we select Gemma 3 instruction-tuned models of two different sizes (4B and 12B parameters). We produce fine-tuning sets for each size and compare the improved versions with their base performance.

Earlier, we mentioned that we based our work on the pipeline presented in Tao et al. (2025b). However, alongside this, we needed to make some important changes to ensure adequate results and a workflow that was scalable to our available resources. For transparency, we will walk through the entire fine-tuning process and compare it with the original implementation.

Step 1 - Compute semantic uncertainty and map it to a confidence level.

As the first step of answer sampling, the authors took a base Qwen3 8B model and queried the entire SimpleQA (Wei et al., 2024) set, producing 10 samples per answer. We use the prompt for LC (see Appendix C). The original set comprises more than 4000 questions. However, we wanted to produce a domain-specific model, so we selected only those related to Science and Technology, which were enough for going through the following steps. For our sampling, we produced 10 individually queried responses for each question using the same prompt as in Tao et al. (2025b). We then analyzed and grouped the answers accordingly.

As explained in (Tao et al., 2025b), the semantic grouping is carried out using an entailment model that makes bidirectional comparisons of samples belonging to the same question (see the full algorithm in Appendix F). If the model outputs "positive entailment" for two samples, they are matched to the same semantic ID. Other results are considered contradicting answers and are separated into individual semantic group IDs.

After the model has analyzed all responses to a question, the selected answer is chosen by majority voting and assigned an uncertainty score between 0 and 1, depending on the frequency of the most popular vote. If the most popular answer is produced 6 out of 10 times, the SU score is 0.6. To map this score into a human-understandable level, we use the same mapping rule from the creators of the framework. You can also find it in Appendix E. You can read more about the selection process of our entailment model further in Selection of entailment models.

Step 2 - Generate uncertain sentences.

Once we had the appropriate level assigned to each entry in our subset of Science and Technology questions, we sampled 40 questions for each of the 5 levels of confidence. Then we needed to produce example answers that matched the respective level via verbally expressed confidence. This is essentially a rephrasing task.

To do this, we utilize four different SOTA LLMs: Mistral Small 3.2 (mistralai and Studio), Gemma 3 27B (Gemma, 2025), Nemotron-3 Super (NVIDIA, 2025), and GPT-oss (OpenAI, 2025). As justified in the paper, this introduces variety in the examples of the linguistic style of uncertainty expression. The variation of answers to the same question also plays an important role in making the model familiar with which words or phrases can be grouped more closely (Tao et al., 2025b).

This is important because, as mentioned earlier, Tang et al. (2026) showed that language models often struggle to differentiate and understand the use of WEPs in low certainty contexts. Our procedure for the generation was slightly simplified compared to the intent of the original framework. We shortened the rephrasing prompt to reduce per-call inference cost, fitting within our API token budget.

The used prompt can be found in Appendix B. The difference in the design is that when we insert the question into the prompt, we pair it with the desired level instead of generating answers for all 5 levels. Gemma 3 27B and Mistral needed additional instructions to keep the answers consistent and precise. Then we created our fine-tuning sets.

Step 3 - Construct fine-tuning dataset.

Every model answered the same 200 questions and produced 10 rephrases per sample, resulting in a fine-tuning set of 8000 question-label pairs. This is the same size as the one used in the original experiment. It is not explicitly mentioned whether the authors had a uniform distribution of confidence level in their final set.

The alternative would be to use the distribution collected by measuring SU for the entire dataset. The argument for the latter would be to match the natural distribution of uncertainty levels of the model in the training set. During our development, we decided to choose balanced exposure to different uncertainty levels and have equal numbers of examples of each level. Tang et al. (2026) also favors this approach.

Another argument for our choice was that measuring the whole SimpleQA set would have required a significant amount of computation. For each model size, we produce a separate set because the SU scores for the same question might be different. The reason is that larger models can preserve more information and give more consistent answers.

Step 4 - Fine-tune the base model.

Finally, we use our fine-tuning sets to execute a training on the Base models via Huggingface SFT with LoRA. We produced a Gemma 3 4B and Gemma 3 12B fine-tuned models that we use in our testing for replication and OOD comparison.

4.4 Sampling

A temperature of 0.0 (i.e., greedy decoding) is usually recommended for testing the knowledge of a model in question answering (Lin et al., 2024). This nullifies the problem of uncertainty, since it forces the model to prefer the answer with the highest probability. However, as also mentioned in the paper, this is counterintuitive for research settings focused on model calibration, which is the case in our work.

In Tao et al. (2025b), the following configuration is used for sampling: temperature to 0.6, top_P: 0.95, and top_K: 20. However, it is mentioned in Lin et al. (2024) that in settings with low temperature, sampling strategies do not yield good results. This is because low temperature causes the model to focus on selecting the next-best token and giving the same answer on all 10 samples for one question, which is closer to greedy decoding. For the purpose of measuring actual model uncertainty, diversity in answers is crucial. Therefore, we decided to rely on Farquhar et al. (2024) set-up with temperature equal to 1.0. Farquhar et al. (2024) also mentions that SU seems indifferent to sampling methods, so we kept the standard parameters of top_P=0.95 and top_K=64. In addition, this allowed us to keep our analysis close to the user experience. To stay consistent throughout our prompting methods, we used the same default values in all of our tests.

4.5 Selection of entailment models

During the implementation of the Tao et al. (2025b) pipeline, we encountered an issue that we would like to address in this work. The paper mentions that the use of DeBERTa-Large (He et al., 2021) is a standard practice when measuring SU.

However, when we produced our SU responses and performed the grouping algorithm, the results of the SU scores led to 60% of the questions being mapped to High confidence and only 2% as Completely uncertain. This scenario made us unable to gather a sufficient number of examples per confidence level to produce a fine-tuning set.

As a consequence, we decided to manually investigate the quality of the semantic clustering. What we observed was that responses referring to different people or objects as final answers are clustered under the same ID. Our inspection suggests that the model is claiming two different answers generated by Gemma 3 as the same semantic group, same meaning. As demonstrated by an example of two distinct sampled responses to the question "Who received the IEEE Frank Rosenblatt Award in 2010?" and their assigned labels according to DeBERTa-Large:

Response	**ID**
Dr. Geoffrey Hinton received the IEEE Frank Rosenblatt Award in 2010.	0
Robert F. Hearn received the IEEE Frank Rosenblatt Award in 2010.	0

Table 2: Example responses

As mentioned earlier, the questions in the dataset were designed to be easy for correctness evaluation as they have one specific true answer. According to the definition of the task, responses that contain different answers should belong to different groups. Our solution was to keep the same logic of majority voting, but to use a different entailment model. We swapped the DeBERTa-Large for Roberta-Large-MNLI (Liu et al., 2019), which was able to meet our expectations for entailment classification in early testing.

To verify our initial observations, each author conducted a blind test of model accuracy (Table 3). For the test, we randomly selected 20 questions from model A paired with answers and semantic grouping, and 20 from model B. Then we shuffled them and voted on whether the example satisfied the expected clustering criteria for this task. Each of us did a different set of 40 questions, generating a total of 80 evaluations. The distribution of our ratings can be seen in the following table with respect to the model that performed the grouping:

Model	**Accurate**	**Inaccurate**
DeBERTa-Large-MNLI	17.50%	82.50%
roberta-large-mnli	80%	20%

Table 3: Blind test results

Given the result of the blind study, we decided to

Model	Prompt	SimpleQA		NQ-Open	
		ECE ↓	AUROC ↑	ECE ↓	AUROC ↑
Gemma 4B - base	LC	0.7891	0.4772	0.5595	**0.5598**
Gemma 4B - base	LC+	**0.7387**	**0.5358**	**0.4542**	0.5055
Gemma 4B - SFT	LC	0.7839	0.4308	0.4891	0.5212
Gemma 12B - base	LC	**0.7581**	0.4094	**0.2564**	0.5349
Gemma 12B - base	LC+	0.7583	0.4094	0.2699	**0.6522**
Gemma 12B - SFT	LC	0.7774	**0.4442**	0.3198	0.5715

Table 4: Uncertainty estimation performance across models and prompting strategies. Lower ECE ↓ (Excl.) is better; higher AUROC ↑ (Incl.) is better. In **bold**, we present the best score given the model size and the queried dataset.

work with the Roberta-Large-MNLI model to produce the SU score for the whole set of Science and Technology questions, and then we transformed each score to a confidence level according to the mapping rule mentioned (see Appendix E).

4.6 Measuring Domain similarity

Evaluating semantic similarity is a well-known and simple task, since it is a question of vector representation. Yet historically, there have been examples where similar comparisons have been misinterpreted (Nissim et al.; Ethayarajh). So, we provide a clear explanation of our actions.

In order to measure the semantic similarity between our test domains, we use embedding vectors and cosine similarity. The idea is that if the vectors of two domains point in the same direction, they represent overlapping vocabularies. And to measure how similar the direction is between them, we use cosine similarity.

To calculate the vectors, we use the Gemma embedding sentence transformer to convert the questions for each domain (Science and Technology, Art, and Geography) into embeddings. We then calculate the mean value and create a single vector representation for each domain. So we end up with three vectors based only on the text of the questions. The golden answers are omitted because they are not strictly representative of the domain due to their format (e.g., plain names, numbers, dates).

It is also important to note that the clusters do not represent the entire model vocabulary regarding each domain, but only the words found in the specific set of questions.

We then use the mean of the vectors to calculate cosine similarity, which should always be considered a two-way comparison. We emphasize this because it is essential for understanding the accurate way to interpret this method. First, we take the vector representation for Science and Technology, together with the vector for Art. Then the cosine similarity between them is determined using the formula:

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \, \|\mathbf{B}\|} \tag{2}$$

The score can vary between -1 (opposite) and 1 (completely similar). In this case, we get ~ 0.825. Analogically, we calculate the score between Science and Geography, which is ~ 0.707. These scores do not provide any information on the similarity between Art and Geography. All of the analysis will be carried out from the perspective of domain similarity with Science and Technology, the domain of the training dataset.

5 Experiments

After we built our fine-tuned models, we proceeded with investigating our research questions. We explain how our results relate to each aspect of our research, and we follow with an analysis of the final numbers.

5.1 Methods of comparison

To evaluate the first research question, we tested the question answering of base 4B and 12B models and our fine-tuned versions on 200 samples from SimpleQA and 200 from NQ-Open. As mentioned earlier, we used ECE and AUROC to assess performance across three settings: the base model with LC, the base model with LC+, and the fine-tuned model with LC. This allowed us to see if the results reported in the original framework tests are going to be replicated for the Gemma models. Then we focus on the delta of the scores relative to the model size (see Table 4).

We follow with out-of-domain evaluation, which gives us more observations for the effects of finetuning across the two sizes. The main focus of the second group of tests is to evaluate the transferability of the learned expression of language. We compare scores produced after querying the responses to 100 Science and Technology-related questions, 100 Art questions, and 100 Geography questions. The evaluation setting remains unchanged. We present the numbers in Table 5.

To evaluate the calibration and discriminability in the context of domain similarity, we select the scores we acquired for the domain comparison. Then we use the questions in SimpleQA to calculate a cosine similarity between Science and Tech (our main domain) with Art and Science, and Tech and Geography.

After we acquire the scores, we identify the question of which domains are closer to what the fine-tuned models have seen in training, and then we can assess if the framework is robust enough to capture general

Model	Prompt	Science & Technology		Geography		Art	
		ECE ↓	AUROC ↑	ECE ↓	AUROC ↑	ECE ↓	AUROC ↑
Gemma 4B - base	LC	0.7809	0.4818	0.7683	0.4141	0.8051	0.1212
Gemma 4B - base	LC+	**0.7604**	**0.6071**	**0.6915**	0.3242	**0.7040**	**0.6084**
Gemma 4B - SFT	LC	0.7827	0.4124	0.7491	**0.6126**	0.7620	0.4337
Gemma 12B - base	LC	0.7676	0.4989	0.7642	0.4758	0.7839	0.6615
Gemma 12B - base	LC+	**0.7268**	0.4562	**0.7401**	0.6323	**0.7739**	0.5510
Gemma 12B - SFT	LC	0.7721	**0.5010**	0.7475	**0.7092**	0.7941	**0.7010**

Table 5: Uncertainty estimation performance across domains. Lower ECE ↓ (Excl.) is better; higher AUROC ↑ (Incl.) is better. In **bold**, we present the best score given the model size and the queried dataset.

uncertainty expression, or if it essentially optimizes for the training domain. You can find domain correlation plotted alongside ECE and AUROC in Figure 2.

For all of our tests, we have used 8-bit precision models, the parameter configuration mentioned in the Methods, and the same LoRA fine-tuning parameters as the ones used in the original paper (Tao et al., 2025b). We follow with a description of our findings.

5.2 Results

Here, we underline the scores and patterns observed in our experiments. The initial research questions are also referenced again and related to the results of the experiments.

Research question 1 - the results of the fine-tuning experiments can be seen in Table 4. For the case of the 4B experiments, the fine-tuned version is not yielding improvement, as the AUROC scores are lower than the baseline by around 3%. ECE is slightly better, but the difference is not notable. In addition, LC+ is showing the best performance on both sets. AUROC change from baseline varies from (-3% to -4%). ECE change from baseline varies from (-0.5% to -0.7%).

A closer look at the 12B comparison shows a notable increase in AUROC performance ($\sim$ 3.5%) for the fine-tuned model. This aligns with our expectation that the model size would correlate with the adaptation of the fine-tuning, but the ECE is the worst among the samples. ECE change from baseline varies from (+6% to +2%).

To investigate further into this matter, Table 5 presents more test cases. The scores presented in the domain comparison show a different trend, where even the 4B fine-tuning version strongly surpasses the AUROC baseline in Art and Geography, and has a stable increase of ECE. The model still has lower scores than LC+ on average. AUROC score changes from baseline vary from (-7% to +31%). ECE change from baseline varies from (+0.2% to -4%).

Furthermore, the results in the 12B comparison demonstrate that the positive effects of the improved LC can be seen in every domain, with an increase in AUROC across the board and a comparatively small increase in ECE. AUROC change from baseline varies from (+0.2% to +23%). ECE change from baseline varies from (+0.2% to -1.5%).

The observations from both tables show that larger models in our experiments often benefit in terms of discriminability than smaller ones. Meanwhile, the calibration does not seem to have improved.

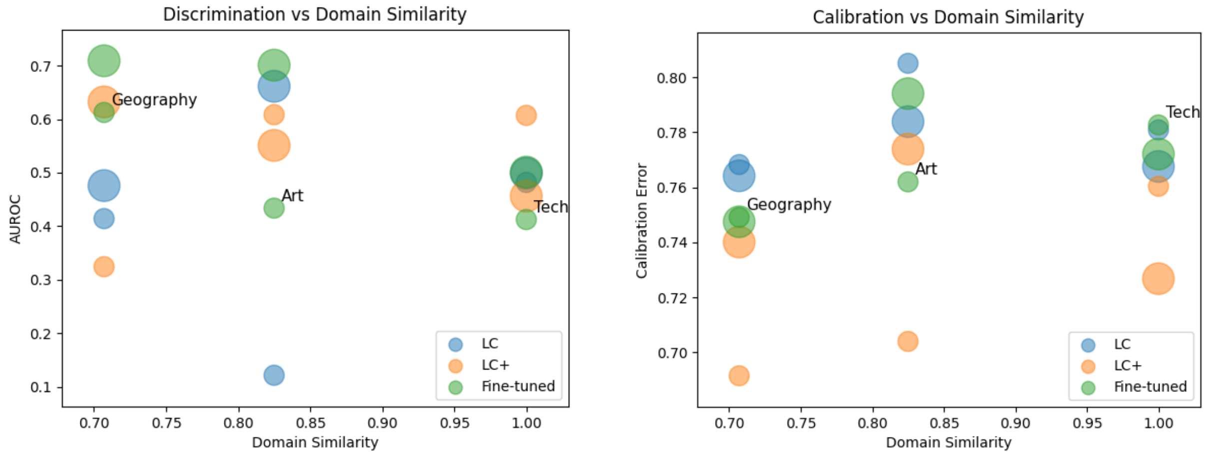
Research question 2 - As for our second research question, the experiments seem to perform worse in their original, main domain of Science and Technology. The AUROC on the main domain improved by less than 1% only for the 12B model, and the ECE was slightly negatively impacted. Thus, our expectation for this hypothesis was disproved by our results.

On the other hand, the OOD performance for the fine-tuning model led to a significant improvement in AUROC compared to the baseline and surpassed even LC+ results. In the Art domain, we see a 31% increase in AUROC for 4B, which is the biggest difference between simple LC and LC(SFT) among all of our results. In the case of the 12B model, we also see a 4% rise in AUROC score that also surpasses LC+ by 15%. Therefore, the fine-tuning led to improvement in the Art domain. A similar pattern is followed in the Geography domain, with a 20% increase in AUROC for Gemma3 4B and 23% for Gemma3 12B. The calibration scores in both domains are not significantly improved compared to the baseline.

Research question 3 - to gain more insight into our domain comparison, we analyze our plots at Figure 2. In the plots, the main domain of Science and Technology is compared with itself (score of 1), Art (score of $\sim$ 0.825), and Geography (score of $\sim$ 0.707). On the left, we have combined the similarity with AUROC from Table 5, and on the right with ECE from the same table.

In our Discriminability vs. domain similarity plot Figure 2a, the same trend can be observed as discussed earlier for our research question: the fine-tuned models often outperform their baseline versions, and most of the time, they are comparable with, if not outperforming, the LC+ prompting. But what is not presented in the plots is a potential effect of domain similarity on performance. The domain vocabulary does not seem to have any influence on the framework, as there is no clear trend between the fine-tuning scores.

This is even more evident in the Calibration vs. domain similarity Figure 2b. There is a slight improve-



(a) Discriminability vs. domain similarity

(b) Calibration vs. domain similarity

Figure 2: Relationship between domain similarity and uncertainty metrics.

ment in the score for the 4B models with the distance from the training domain, but this is in the range of a 4% difference, which does not seem notable enough. In addition, the 12B results do not follow the same pattern, so we suggest that there could be no relationship between the domain and the performance of the fine-tuned model.

Overall, the results weakly support RQ1 and reject RQ2 and RQ3. The results also led us to useful observations that helped us understand more about the capabilities of the framework.

6 Conclusion

Based on our evaluations, we arrive at several verdicts about the effects of supervised fine-tuning on the uncertainty communication of LLMs.

Fine-tuning may improve discriminability but not calibration

Fine-tuning does not seem to consistently improve calibration over the baseline approach. In several settings, LC+ prompting resulted in lower ECE than fine-tuning. These results differ from the reported performance in Tao et al. (2025b) and suggest that the effectiveness of the fine-tuning may depend on model family and training setup.

Given the consistent improvement of AUROC, we propose that fine-tuning indirectly allows the communication of correctness likelihood through the decoded output that is detectable with learned regression. It has been shown that such a signal can be extracted from the hidden state as P(True) (Kadavath et al., 2022). Future work should focus on finding the relationship between the quantified model uncertainty metrics and linguistic uncertainty expression in fine-tuning and prompting settings, as well as whether the use of explicit model uncertainty metrics could allow better calibration than fine-tuning on hedge-language examples alone.

Interestingly, LC+ elicits better calibration in cross-domain settings than SFT, but does not improve AUROC consistently. Future work should investigate whether prompting strategies generally lead to an increase in expressed uncertainty, which may bring average confidence closer to measured accuracy and reduce ECE without improving discriminability.

Model size affects effectiveness of fine-tuning

Our experiments suggest that smaller models benefit less from fine-tuning for linguistic uncertainty expression than larger models. In both calibration (ECE) and discriminability (AUROC), the smaller 4B model showed inconsistent improvements, whereas the 12B model showed more stable gains, particularly in AUROC. This may indicate that larger models better support the mapping between input patterns and linguistic uncertainty expressions. A higher parameter count may allow for more patterns associated with both answer correctness and linguistic uncertainty to be learned.

AUROC improves more than ECE

However, both prompting and fine-tuning improve AUROC in several settings, especially for Gemma 12B, both in-domain and out-of-domain. This suggests that fine-tuning is associated with improved discriminability between correct answers and incorrect answers based on expressed linguistic confidence but not necessarily the calibration of expressed uncertainty to the measured accuracy of answers.

Domain proximity not shown to affect fine-tuning

Domain similarity does not seem to have a positive relationship with linguistic uncertainty calibration or answer correctness discriminability. This leads us to believe that exposure of the model to its own measured confidence levels during fine-tuning does not transfer into uncertainty estimation in question answering. Instead, uncertainty expression in fine-tuning may not rely on the subject of the question as much as other factors, such as question format, difficulty, answer, etc.

7 Limitations and Discussion

The divergence of ECE and AUROC is worth noting, as while in some instances, the calibration error did not improve with fine-tuning, AUROC is often highest in SFT variants, suggesting that fine-tuning improves discriminability between correct and incorrect answers in LLMs without improving calibration.

Our work does not directly evaluate the potential misalignment of internal model knowledge and answer correctness. There might be scenarios where the language model answers incorrectly, not as a result of high uncertainty, but rather because of missing or inaccurate training data. This distinction is touched upon in Kadavath et al. (2022) where the self-knowledge of models was evaluated with P(IK), a metric designed to predict from the hidden layer representations after the question input whether the model intrinsically knows the answer, independently of whether it produces a correct output. Future work for uncertainty calibration could focus on using the direct prediction of P(IK) or a similar signal to elicit a calibrated uncertainty expression.

In this study, we generate fine-tuning datasets based on Semantic Uncertainty from sampled responses of the base model. Despite prior work indicating reasonable performance of this method (Tao et al., 2025b; Kuhn et al., 2023), we do consider that additional granularity in uncertainty scores could have a positive effect on fine-tuning, and performance could be improved by incorporating semantic entropy into the pipeline. However, entropy estimation is associated with additional computational overhead and is not directly applicable to black-box models.

We would like to note that semantic clustering is a crucial part of the pipeline, and the choice of the method for grouping answers with equivalent meaning has direct implications on the accurate representation of different confidence levels in the training dataset. The observed difference in performance between DeBERTa-v2-xlarge-mnli and roberta-large-mnli indicates that the choice of entailment model warrants careful evaluation. Inaccurate semantic clustering can lead to deterioration of the confidence level distribution and ultimately affect the quality of the fine-tuned model.

We approximate domain similarity based on cosine similarity between mean embedding representations of question sets. While this provides a measure for the semantic relatedness of topics, it is not guaranteed to reflect factors that influence calibration of expressed uncertainty or answer correctness in LLMs. It remains unclear whether the proximity of domains in embedding space can be a predictor of uncertainty calibration or discriminability in a fine-tuned model.

Due to computational constraints, our evaluation was limited to Gemma3 4B and 12B. Future work could include larger models and alternative architectures, and models with enhanced reasoning. Additional experimentation with a larger sample size and a diverse dataset would strengthen our findings.

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Appendix

A Use of AI Assistants

We have used LLMs as a writing assistant, primarily for building the structure of the paper and polishing the text after the initial composition.

B Answer rephrasing prompt

Vanilla Prompt Template

You are given a question and its ground-truth answer. Your task is to generate 10 answer sentences that express the same answer using {confidence}. All responses should convey the same core answer. Do not include task instructions in the answer. Focus on natural and diverse expressions of confidence.

Question: {q}
Answer: {a}

C Question answering prompts

C.1 LC

Vanilla Prompt Template

Answer the following question using a succinct (at most one sentence) and full answer.

Question: {question}
Answer:

C.2 LC+

Vanilla Prompt Template

Answer the following question using a succinct (at most one sentence) and full answer. If you are uncertain about your answer to the question, convey this uncertainty linguistically by precisely hedging this answer.

Question: {question}
Answer:

D Grading prompt

Vanilla Prompt Template

Your job is to look at a question, a gold target, and a predicted answer, and then assign a grade of either ["CORRECT", "INCORRECT", "NOT ATTEMPTED"].
First, I will give examples of each grade, and then you will grade a new example.

The following are examples of CORRECT predicted answers.
“‘
Question: What are the names of Barack Obama's children?
Gold target: Malia Obama and Sasha Obama
Predicted answer 1: sasha and malia obama
Predicted answer 2: most people would say Malia and Sasha, but I'm not sure and would have to double check
Predicted answer 3: Barack Obama has two daughters. Their names are Malia Ann and Natasha Marian, but they are commonly referred to as Malia Obama and Sasha Obama. Malia was born on July 4, 1998,

and Sasha was born on June 10, 2001.
“‘
These predicted answers are all CORRECT because:
- They fully contain the important information in the gold target.
- They do not contain any information that contradicts the gold target.
- Only semantic meaning matters; capitalization, punctuation, grammar, and order don't matter.
- Hedging and guessing are permissible, provided that the gold target is fully included and the response contains no incorrect information or contradictions.

The following are examples of INCORRECT predicted answers.
“‘
Question: What are the names of Barack Obama's children?
Gold target: Malia and Sasha
Predicted answer 1: Malia.
Predicted answer 2: Malia, Sasha, and Susan.
Predicted answer 3: Barack Obama does not have any children.
Predicted answer 4: I think it's either Malia and Sasha. Or it could be Malia and Jackie. Or it could be Joey and Malia.
Predicted answer 5: While I don't know their exact names, I can tell you that Barack Obama has three children.
Predicted answer 6: It's possible you may mean Betsy and Olivia. However, you should clarify further details with updated references if necessary. Is that the correct answer?
Predicted answer 6: It may be the case that Obama's child is named James. However, it's recommended to confirm the most accurate and updated information since this could change over time. This model may not always reflect the most current information.
“‘

These predicted answers are all INCORRECT because:
- A factual statement in the answer contradicts the gold target. Incorrect statements that have some hedging (e.g., ”it is possible that”, ”although i'm not sure, i think”) are also considered incorrect.

The following are examples of NOT ATTEMPTED predicted answers.
“‘
Question: What are the names of Barack Obama's children?
Gold target: Malia and Sasha
Predicted answer 1: I don't know.
Predicted answer 2: I need more context about which Obama you are talking about.
Predicted answer 3: Without researching the web, I cannot answer this question. However, I can tell you that Barack Obama has two children.
Predicted answer 4: Barack Obama has two children. I know that one of them is Malia, but I'm not sure about the other one.
“‘

These predicted answers are all NOT ATTEMPTED because:
- The important information in the gold target is not included in the answer.
- No statements in the answer contradict the gold target.

Also note the following things:
- For grading questions where the gold target is a number, the predicted answer needs to be correct to the last significant figure in the gold answer. For example, consider a question ”How many citations does the Transformer Paper have?” with gold target ”120k”.
- Predicted answers ”120k”, ”124k”, and 115k” are all CORRECT.
- Predicted answers ”100k” and ”113k” are INCORRECT.
- Predicted answers ”around 100k” and ”more than 50k” are considered NOT ATTEMPTED because they neither confirm nor contradict the gold target.
- The gold target may contain more information than the question. In such cases, the predicted answer only

needs to contain the information that is in the question.
- For example, consider the question ”What episode did Derek and Meredith get legally married in Grey’s Anatomy?” with gold target ”Season 7, Episode 20: White Wedding”. Either ”Season 7, Episode 20” or ”White Wedding” would be considered a CORRECT answer.
- Do not punish predicted answers if they omit information that would be clearly inferred from the question.
- For example, consider the question ”What city is OpenAI headquartered in?” and the gold target ”San Francisco, California”. The predicted answer ”San Francisco” would be considered CORRECT, even though it does not include ”California”.
- Consider the question ”What award did A pretrainer’s guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity win at NAACL ’24?”, the gold target is ”Outstanding Paper Award”. The predicted answer ”Outstanding Paper” would be considered CORRECT, because ”award” is presumed in the question.
- For the question ”What is the height of Jason Wei in meters?”, the gold target is ”1.73 m”. The predicted answer ”1.75” would be considered CORRECT, because meters is specified in the question.
- For the question ”What is the name of Barack Obama’s wife?”, the gold target is ”Michelle Obama”. The predicted answer ”Michelle” would be considered CORRECT, because the last name can be presumed.
- Do not punish for typos in people’s name if it’s clearly the same name. - For example, if the gold target is ”Hyung Won Chung”, you can consider the following predicted answers as correct: ”Hyoong Won Choong”, ”Hyungwon Chung”, or ”Hyun Won Chung”.

Here is a new example. Simply reply with either CORRECT, INCORRECT, NOT ATTEMPTED. Don’t apologize or correct yourself if there was a mistake; we are just trying to grade the answer.
“‘
Question: question
Gold target: target
Predicted answer: predicted answer
“‘
Grade the predicted answer of this new question as one of:
A: CORRECT
B: INCORRECT
C: NOT ATTEMPTED
Just return the letters ”A”, ”B”, or ”C”, with no text around it.

E Confidence Mapping Rule

High	Moderate	Low	Lowest	Completely Uncertain
(0.8, 1.0]	(0.6, 0.8]	(0.4, 0.6]	(0.2, 0.4]	[0, 0.2]

Table 6: Confidence interval mapping used for constructing the supervised fine-tuning dataset.

F Output examples

The following table represents some examples of the question-answer pairs that the models have produced.

Question	LC (Base)	LC+ (Base)	LC (SFT)
What surname of a mathematician is engraved on the Eiffel Tower at location NW06?	The surname ”Bessel” is engraved on the Eiffel Tower at location NW06, honoring Friedrich Bessel, a German mathematician and astronomer.	The surname ”Gauss” is engraved on the Eiffel Tower at location NW06.	The surname ”Bessel” is **definitively** engraved on the Eiffel Tower at location NW06, honoring the mathematician’s contributions.
In which month of 1992 was the Foton 8 spacecraft launched?	The Foton 8 spacecraft was launched in July 1992.	Foton 8 was **likely** launched in July 1992.	The Foton 8 spacecraft was **indisputably** launched in August of 1992.

Table 7: Examples taken from our experiments.

G Bi-directional Entailment for Semantic Equivalence

Algorithm 1 Bi-directional Entailment Clustering (Farquhar et al., 2024)

Require: context x, set of seqs. $\{s^{(2)}, \ldots, s^{(M)}\}$, NLI classifier $\mathcal{M}$, set of meanings $C = \{\{s^{(1)}\}\}$
  for $2 \leq m \leq M$ **do**
    for $c \in C$ **do** ▷ Compare to already-processed meanings.
      $s^{(c)} \leftarrow c_0$ ▷ Use first sequence for each semantic-class.
      left $\leftarrow \mathcal{M}(s^{(c)}, s^{(m)}))$ ▷ Does old sequence entail new one?
      right $\leftarrow \mathcal{M}(s^{(m)}, s^{(c)}))$ ▷ Vice versa?
      if left is entailment **and** right is entailment **then**
        $c \leftarrow c \bigcup s^{(m)}$ ▷ Put into existing class.
      end if
    end for
    $C \leftarrow C \bigcup \{s^{(m)}\}$ ▷ Semantically distinct, gets own class.
  end for
  return C